

Descriptor: *Comprehensive IEEE Research Data Collections (CIRDC)*

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ABSTRACT The IEEE Xplore database is vital in democratizing access to high-quality research datasets, fostering global collaboration, and promoting interdisciplinary studies. Insights from the IEEE Xplore database support applications in academic collaboration networks, predictive research trends, recommendation systems, and the evolution of scientific discourse. It is downloaded using web data mining methods, such as HTTP requests, web scraping with Selenium, and LXML parsing with BeautifulSoup. These various methods are discussed for their efficiency and complexity. As a means of ensuring the quality of these datasets, we propose the use of cross-repository validation. Source codes and scripts for data collection are provided to promote transparency and reproducibility.

IEEE SOCIETY/COUNCIL Computer Society (CS)

DATA TYPE/LOCATION Text; Worldwide

DATA DOI/PID 10.21227/6514-ay49

INDEX TERMS Academic research, computer science, electrical engineering, IEEE Xplore, open-source.

BACKGROUND

Publishing with IEEE comes with multiple benefits (Fig. 1). In particular, academic sharing of bibliographic references [1] from the IEEE Xplore database [2] plays a crucial role in democratizing access to high-quality datasets for academic research [3]. IEEE Xplore [2] is not only a repository of valuable research data but also a platform for global collaboration between scholars. By providing researchers with a rich, diverse, and easily accessible body of data, IEEE Xplore has the potential to significantly accelerate scientific discovery.

The IEEE Xplore database is instrumental in promoting interdisciplinary research. By providing access to a broad spectrum of data in multiple fields of study, researchers can draw connections between disparate areas of knowledge, leading to innovative breakthroughs and new research directions. The database's comprehensive coverage and ease of access make it an invaluable resource for researchers aiming to stay at the forefront of their fields. However,

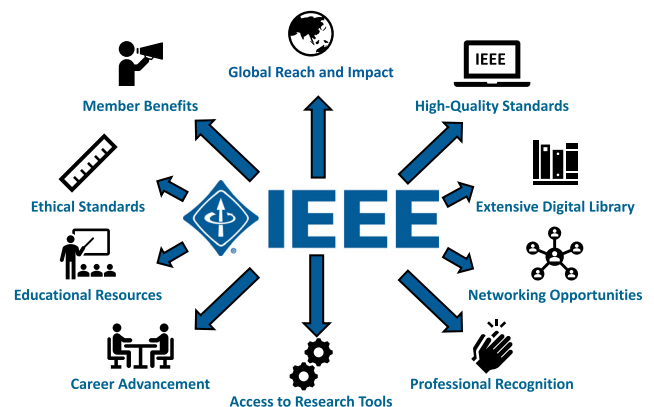


FIG. 1. Benefits of publishing in the IEEE Communities.

there are several limitations in the current functionality of the IEEE Xplore platform, posing challenges for large-scale data collection and advanced data mining efforts.

TABLE I. Type of Github Projects With the List of Their Library Dependencies

Github Project	Mining Libraries	Parsing Libraries	Pros	Cons
IEEE-Xplore Web Crawler [4]	Requests [5]	Regex, Json	Fast and simple for static content	Limited scalability for large-scale scraping
IEEE Xplore-Crawler [6]	Selenium [7]	Regex, Json, Pandas, BeautifulSoup [8]	Handles dynamic content, automates browser actions	Slower, need to provide conference ID
IEEE Crawler [9]	Requests [5]	BeautifulSoup [8], lxml [10], MongoDB	Handles large datasets well	Complex setup and infrastructure
ACM-IEEE-arXiv Info Spider [11]	Scrapy [12], Selenium [7]	MySQL	Support more libraries (ACM, arXiv, AAAI)	Complex setup and infrastructure
IEEEPaperDownloader [13]	urllib2 [5]	Regex	Lightweight source code	Limited information, need to provide paper ID

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headers = {
    'Accept': 'application/json,text/plain,*/*',
    'Accept-Encoding': 'gzip,deflate,br',
    'Accept-Language': 'zh-CN,zh;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6',
    'Connection': 'keep-alive',
    'Content-Length': '147',
    'Content-Type': 'application/json',
    'Referer': 'https://ieeexplore.ieee.org/search/searchresult.jsp?newsearch=true',
    'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) '
                'AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.0.0 '
                'Safari/537.36 Edg/108.0.1462.46',
}

```

FIG. 2. Code Snippet of the headers for HTTP request to the IEEE Xplore search API.

- 1) *Manual Data Querying*: The manual nature of data querying on the IEEE Xplore website can be cumbersome when attempting to retrieve large volumes of academic paper, which is time-consuming and inefficient for researchers seeking to conduct large-scale analysis.
- 2) *Limited Data Flexibility*: The displayed information from IEEE Xplore is fixed and limited by the front-end user interface, lacking the flexibility needed to directly support further information extraction or analysis, e.g., academic collaboration networks, research trend evolution, and predictive analytics.
- 3) *Limited Customized Searches*: While IEEE Xplore provides various search options, researchers are limited when they require more advanced customized searches, e.g., regular expressions, limiting the platform's effectiveness for more in-depth research needs.

To address the above challenges, this article presents a Comprehensive IEEE Research Data Collections, CIRDC. CIRDC includes metadata for all IEEE papers up to July 2024, offering a complete historical overview of published research. We also provide code to facilitate future updates to the database, ensuring that researchers have access to the latest publications as they become available.

The rest of the article is organized as follows. Section “Collection Methods and Design” explores various methods for extracting data from the IEEE Xplore digital library

to create a comprehensive research paper database. Section “Validation and Quality” addresses the validation and quality assurance of the collected data, proposing cross-repository validation techniques. Section “Records and Storage” details the organization and storage of the dataset for easy access and use by researchers. Section “Insights and Notes” discusses the use of the datasets from the IEEE Xplore digital library for collaboration networks, trend prediction, recommendation systems, and discourse evolution.

COLLECTION METHODS AND DESIGN

There are several methods [4], [6], [9], [11], [13] to extract data from the IEEE Xplore digital library to obtain a comprehensive and reliable research paper database. Table I compares their advantages, disadvantages, and library dependencies. Most of these methods use Python and a variety of libraries, including HTTP requests, web scraping with Selenium, and parsing with BeautifulSoup and lxml [8], [10], [14]. Each method has its advantages in terms of speed, efficiency, and complexity.

Data Mining

One common method for data collection is using HTTP requests facilitated by the “requests” and “urllib3” libraries [5]. These libraries provide a straightforward way to send HTTP requests and receive responses with specific request headers and parameters (Fig. 2) to the IEEE Xplore search

API¹ through the representational state transfer (REST) API, allowing programmatic access and retrieval of webpage content. This method is relatively simple to implement and is effective for extracting text-based data and metadata from the HTML content of web pages. However, it may face limitations when dealing with dynamic content or pages that require JavaScript execution.

To address the limitations of HTTP requests, web scraping techniques using the Selenium library [14] are often employed as a way to automate browser actions. Web scraping allows you to interact with various web elements, making it especially useful when it comes to complex scraping tasks, such as handling pagination, clicking on buttons, and filling out forms. The tradeoff is that Selenium-based scraping can be slower and more resource-intensive compared to HTTP request methods [7].

To improve throughput, web crawling and scraping frameworks using the scrapy library [12] allow for the creation of robust spiders that can crawl websites asynchronously and extract data systematically. Its built-in features include handling cookies, sessions, and proxies, further enhancing its capabilities, making it an ideal choice for large-scale data collection projects. To scrape large-scale or sensitive websites, proxy websites are used to avoid IP bans and manage request rates more effectively.

For the proposed CIRDC, we adopt a two-stage strategy to collect the data to circumvent the limitation that the maximum number of entries returned in a single query is restricted to 10000. In the first stage, we query IEEE Xplore to retrieve the publication numbers of all journals and conference proceedings for each year. This enables us to identify the new journals and conferences published in any given year, ensuring no publication is overlooked. In the second stage, we collect the metadata of IEEE papers based on the publication numbers of journals and conferences on a year-by-year basis. As the search results are returned on multiple pages, we handle each page sequentially. The metadata collection process is automated through static HTTP requests, with parameters such as year, page number of search result, and publication number in the dictionary-formatted request headers. The proposed strategy leverages static HTTP requests to maintain the efficiency of the collection process while ensuring that no papers are missed. Meanwhile, it also supports flexible updates to the database for newly introduced journals and conferences.

Data Parsing

Parsing libraries, such as BeautifulSoup [8] and lxml [10], play a crucial role in processing the HTML content retrieved from IEEE Xplore. BeautifulSoup is a widely used library for parsing HTML and XML documents, providing a Pythonic interface to navigate and search the parse tree and allowing us to extract specific elements, attributes, and text content

from web pages with ease [8]. Similarly, lxml is a powerful library for XML and HTML processing, making it an excellent choice for handling large datasets [10]. By combining these parsing libraries with scraping methods, researchers can effectively transform raw HTML data into structured and usable information [15], [16]. As CIRDC is based on the IEEE Xplore Search REST API, it directly retrieves results in JavaScript Object Notation (JSON) format, eliminating the need for complex HTML parsing. After the results are retrieved, we remove redundant metadata fields and reorganize the results according to the publication number and year, ensuring that each paper is properly categorized and easily accessible for further research analysis.

VALIDATION AND QUALITY

Ensuring the technical quality and accuracy of the data collected from the IEEE Xplore database is critical. To achieve this, we propose utilizing multiple cross-repository validation techniques to ensure accuracy in the metadata. We employed two sets of scripts: our custom script and a modified version of the IEEE Xplore-Crawler [6]. Note that our custom script leverages the requests library for efficient HTTP requests and is designed to parse search results based on journals or conferences for each year. In contrast, the IEEE Xplore-Crawler, which relies on the Selenium library, performs data collection by executing browser automation to retrieve search results on a per-paper basis. While their method provides a detailed extraction of individual paper data, it is significantly slower (>100 times) due to the overhead of browser automation and the need to handle each paper separately.

To validate the accuracy of CIRDC, we conducted a comparative analysis using the papers from the proceedings of IEEE Biomedical Circuits and Systems (BioCAS) [17] as a case study. The comparison between our custom script and the IEEE Xplore-Crawler revealed differences primarily in three aspects: date representation, format, and the number of articles collected.

Date Representation

CIRDC uses a shorthand format for publication dates, for example, “19–21 October 2023.” In contrast, the IEEE Xplore-Crawler displays dates in a more verbose format, such as “19–21 October 2023.” This difference in date representation can lead to discrepancies when comparing datasets directly, as the same date might appear differently formatted depending on the source script. However, it is easily resolvable with postprocessing of the dates.

Format Differences

Another difference lies in how article titles are formatted. Our custom script returns titles in a straightforward text format, such as “Highly-Digital 0.0018-mm²/channel Multiplexed Neural Frontend with Time-Based Incremental ADC for In-Brain-Stroke-Cavity LFP Monitoring.” In contrast, the

¹IEEE Xplore Search REST API, <https://ieeexplore.ieee.org/rest/search>

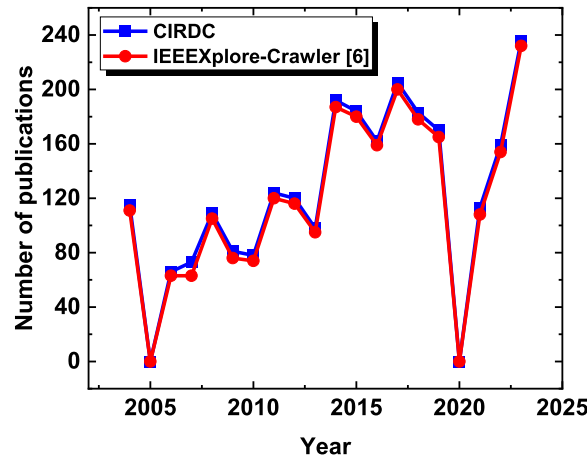


FIG. 3. Retrieved the number of publications in IEEE BioCAS at different years using CIRDC and IEEE Xplore-Crawler [6].

IEEE Xplore-Crawler incorporates HTML tags and special character formatting, resulting in entries such as “Highly-Digital 0.0018-mm²/channel Multiplexed Neural Frontend with Time-Based Incremental ADC for In-Brain-Stroke-Cavity LFP Monitoring.” This inconsistency in formatting can introduce challenges in data processing and integration, requiring additional steps to standardize the datasets for meaningful comparison.

Number of Publications

The number of publications retrieved by each method shows a difference. The IEEE Xplore-Crawler, by mimicking real user interactions within a browser, does not access certain entities that lack hyperlinks, such as cover pages or copyright pages of conference proceedings. As a result, these entities are excluded from the dataset collected by the crawler. In contrast, our custom script returns all entities displayed in the proceedings, ensuring that the number of articles matches the manual counts verified in IEEE Xplore. Fig. 3 presents a comparison of the number of articles retrieved for various years of IEEE BioCAS conference proceedings [17] using the two methods. It is evident that each year, the IEEE Xplore-Crawler consistently retrieves fewer publications compared to CIRDC.

RECORDS AND STORAGE

The IEEE research paper database is organized to facilitate easy access and use by researchers. Our dataset is stored in JSON format as it is easy for humans to read and write and easy for machines to parse. To avoid creating excessively large files and to improve indexing efficiency, each JSON file is saved in a directory named after the “publication number” of the journal or conference. The JSON files are named according to the year of publication. For example, papers from the journal IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I: REGULAR PAPERS published in 2023 would be stored in a file named “2023.json” within the “8919” directory. Each JSON file consists of a list, where each

TABLE II. Structure of JSON Data File

Name	Description	Type
publicationNumber	Identifier for the journal/conference	String
doi	Digital Object Identifier of the paper	String
publicationYear	Year the paper was published	String
publicationDate	Full date of publication	String
articleNumber	A unique number assigned to the paper	String
articleTitle	Title of the paper	String
volume	Volume number	String
issue	Issue number	String
startPage	Starting page number	String
endPage	Ending page number	String
publisher	Name of the publisher	String
articleContentType	Type of the paper (journal, conference, magazine, or early access article)	String
publicationTitle	Name of journal/conference	String
authors	A list of authors	List

TABLE III. Structure of Authors' Field

Name	Description	Type
id	ID number of the author in IEEE system	Int
preferredName	Full name of the author	String
firstName	First name of the author	String
lastName	Last name of the author	String

TABLE IV. Types of Papers in IEEE Xplore [2] as of 20 June 2024

Type	Number of Articles
Conferences	4 591 660
Journals	1 425 264
Magazines	214 106
Books	76 662
Early Access Articles	22 647
Standards	13 746
Total	6 344 708

element in the list represents an entity corresponding to a single paper. The description of each entity is detailed in Table II. Note that “authors” field is a list, where each element in this list contains information about each author, as shown in Table III. The number of articles included in CIRDC is shown in Table IV.

INSIGHTS AND NOTES

The data collected from IEEE Xplore offers a wealth of insights that can be harnessed for various academic and professional applications [3], [15], [18], [19], [20], [21], [22], [23]. In discussing the caveats and special case usages of the CIRDC, it is essential to explore both its inherent limitations and its potential for diverse applications beyond the scope of the original research.

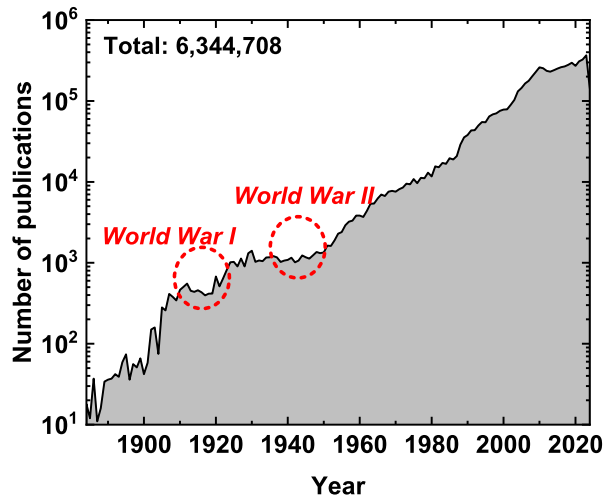


FIG. 4. Number of IEEE Research Papers from 1884 to 2024.

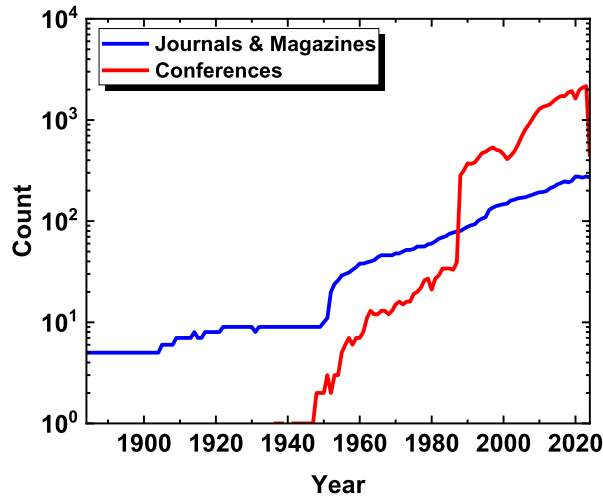


FIG. 5. Number of IEEE Conference and Journals from 1884 to 2024.

Evolution and Impact of IEEE Publications and Conferences from 1884 to 2024

Figs. 4 and 5 depict a substantial increase in IEEE publications from 1884 to 2024, showcasing the organization's expanding impact on technological advancement and research dissemination. Fig. 4 reveals an exponential rise in the number of research papers, especially post-World War II, with significant acceleration during major technological revolutions such as the information technology boom. By 2024, the cumulative number of publications reached 6 344 708. Fig. 5 illustrates the growth of IEEE journals, magazines, and conferences, with a steady increase in journal and magazine counts in the early 20th century and a dramatic surge in conferences from the 1960s onward. This surge aligns with global collaborations and rapid technological advancements. The decentralized “professional group” structure was adopted to incorporate the new fields (e.g., electronic computers and information theory) by holding specialized conferences and publishing specialized journals

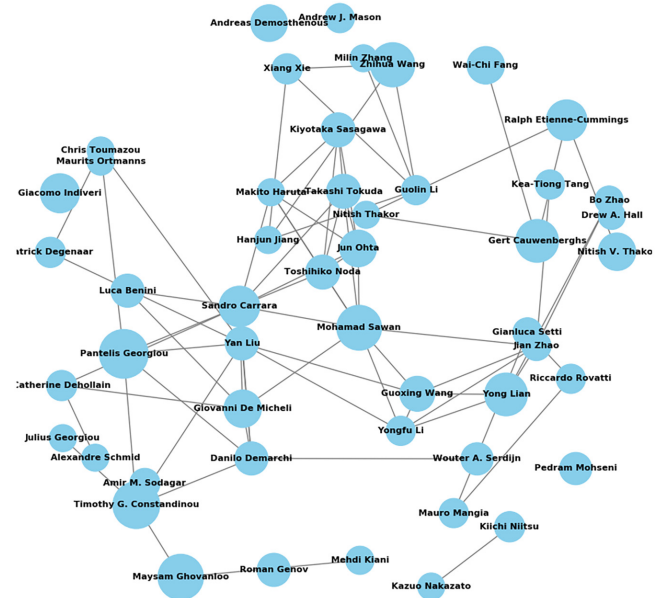


FIG. 6. Visual example of collaboration network in IEEE BioCAS.

from 1948 [24]. The end of the Cold War in the early 1990s further accelerated this surge, as geopolitical tensions eased and international scientific collaboration expanded. Additionally, China's increasing participation in global research and development, along with its rapid technological growth, significantly contributed to the rise in IEEE conference activities [24]. The trends highlight IEEE's essential role in fostering global collaboration, reflecting rapid technological progress and commitment to providing a vast knowledge repository and dynamic forum for innovative ideas.

Enhancing Academic Collaboration Networks

The CIRDC dataset can be utilized to map out collaboration networks among researchers [25]. One can uncover hidden collaborations and influential researchers within specific domains by analyzing co-authorship patterns [20], [22]. Fig. 6 shows a visual example of collaboration network in IEEE BioCAS [17] from 2004 to 2023. Each node represents an author with the node size proportional to the author's publication count (we only select some of the authors with the highest number of publications for simplicity). Connections between nodes indicate co-authorship relationships, allowing us to identify high-frequency collaborators and distinct research teams. Larger nodes indicate prolific authors with extensive co-authorship ties, often serving as central figures in their respective networks. Additionally, the network reveals clusters of closely collaborating authors, reflecting the presence of research groups and key connections bridging different clusters, highlighting the role of influential researchers in knowledge dissemination. However, a caveat to this approach is the potential bias introduced by prolific authors, which may overshadow emerging researchers. Special consideration must be given to normalizing such biases

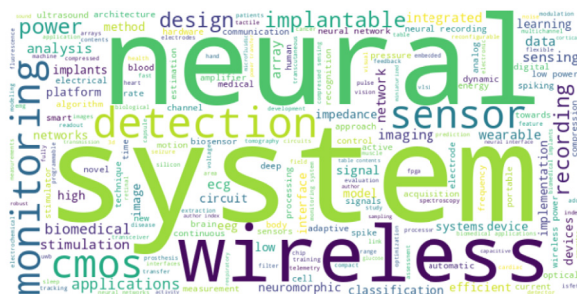


FIG. 7. Wordcloud of paper titles in IEEE BioCAS.

to ensure a more equitable representation of the academic landscape.

Predictive Analysis of Research Hotspots and Trends

The data in CIRDC provide a comprehensive overview of research activities and collaborations over time. By analyzing the keywords from paper titles across different years, we can identify the most frequently discussed topics and their evolution [18]. For instance, Fig. 7 highlights the prominent research keywords such as “neural,” “system,” “wireless,” “sensor,” and “monitoring” in IEEE BioCAS [17]. Such visual representations help in quickly identifying key areas of focus within the research community. Furthermore, leveraging machine learning algorithms, the dataset can predict future research trends by identifying emerging topics and declining areas of interest [25]. This predictive analysis can inform funding agencies and academic institutions about potential areas for investment. The caveat here is the need for sophisticated algorithms capable of discerning genuine trends from transient spikes in interest, often driven by short-term factors such as media coverage and funding announcements. Furthermore, the dataset might not capture all the nuances of interdisciplinary research, leading to potential gaps in trend analysis.

Developing Recommendation Systems for Academic Literature

The dataset can be instrumental in building recommendation systems for academic literature, helping researchers discover relevant papers based on their reading history and research interests [25]. A key challenge is the diversity and volume of data, requiring advanced natural language processing techniques to accurately match papers with user preferences. Another caveat is the potential for echo chambers, where researchers are only exposed to work within their field, missing out on valuable interdisciplinary insights. Thus, it is crucial to design the system to encourage cross-disciplinary recommendations.

Studying the Evolution of Scientific Discourses

The dataset offers a unique opportunity to study how scientific discourses evolve by tracking the development of

key concepts and terminologies within research fields. This can reveal how ideas propagate, transform, and sometimes converge with other disciplines [18]. A caveat here is the challenge of semantic shifts, where the meaning of terms may change over time, complicating longitudinal analysis. Advanced text mining techniques and historical context consideration are necessary to accurately trace these evolutions. Moreover, cultural and regional differences in scientific discourse might require localized analysis to avoid generalizations.

SOURCE CODE AND SCRIPTS

The Github repository [1] at <https://github.com/SJTU-YONGFU-RESEARCH-GRP/CIRDC> is built on the principles of transparency and reproducibility. In addition to the database, it includes source codes for all aspects of the data downloading, parsing, extraction on, and cleaning from IEEE Xplore [2], and the software dependencies required to replicate the database’s operations. It also documents comments that explain its functionality and usage, enabling other researchers to adapt it for their own data collection needs.

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The authors have declared no conflicts of interest.

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